Package ‘manMetaVAR’

November 4, 2025

Title Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients

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BugReports https://github.com/jeksterslab/manMetaVAR/issues

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LazyData true

Roxygen list(markdown = TRUE)

Depends R (>= 4.0.0), OpenMx, fitDTVARMxID, metaVAR

Imports stats, utils, simStateSpace, mlVAR

Remotes jeksterslab/fitDTVARMxID, jeksterslab/metaVAR

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Contents

`coef.manmetavar.dtvar` *Parameter Estimates (FitDTVAR)*

Description

Parameter Estimates (FitDTVAR)

Parameter Estimates (FitMetaVAR)

Parameter Estimates (FitMplus)

Usage

```
## S3 method for class 'manmetavar.dtvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
```

```
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)

## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, ...)
```

Arguments

`object`	Object of class `manmetavar.mplus`.
`alpha`	Logical. If `alpha = TRUE`, include estimates of the `alpha` vector, if available. If `alpha = FALSE`, exclude estimates of the `alpha` vector.
`beta`	Logical. If `beta = TRUE`, include estimates of the `beta` matrix, if available. If `beta = FALSE`, exclude estimates of the `beta` matrix.
`nu`	Logical. If `nu = TRUE`, include estimates of the `nu` vector, if available. If `nu = FALSE`, exclude estimates of the `nu` vector.
`psi`	Logical. If `psi = TRUE`, include estimates of the `psi` matrix, if available. If `psi = FALSE`, exclude estimates of the `psi` matrix.
`theta`	Logical. If `theta = TRUE`, include estimates of the `theta` matrix, if available. If `theta = FALSE`, exclude estimates of the `theta` matrix.
`converged`	Logical. Only include converged cases.
`grad_tol`	Numeric scalar. Tolerance for the maximum absolute gradient if `converged = TRUE`.
`hess_tol`	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if `converged = TRUE`.
`vanishing_theta`	Logical. Test for measurement error variance going to zero if `converged = TRUE`.
`theta_tol`	Numeric. Tolerance for vanishing theta test if `converged` and `theta_tol` are `TRUE`.
`...`	additional arguments.
`median`	Logical. If `median = TRUE`, return median of the posterior. If `median = FALSE`, return mean of the posterior.

Value

Returns a list of vectors of parameter estimates.

Returns a vector of estimated parameters.

Returns a vector of estimated parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

`Compress` *Compress Replication*

Description

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`confint.manmetavar.metavar`
Confidence Intervals for the Parameter Estimates (FitMetaVAR)

Description

Confidence Intervals for the Parameter Estimates (FitMetaVAR)

Confidence Intervals for the Parameter Estimates (FitMplus)

Usage

```
## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, lb = TRUE, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)
```

Arguments

`object`	Object of class `manmetavar.mplus`.
`parm`	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
`level`	the confidence level required.
`lb`	Logical. If `TRUE`, returns profile likelihood-based confidence intervals. If `FALSE`, returns Wald confidence intervals.
`...`	additional arguments.
`burnin`	Integer indicating initial samples to discard. If `burnin = NULL`, do not discard anything.

Value

Returns a matrix of confidence intervals.

Returns a matrix of confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

`Dynamics` *Process Dynamics*

Description

Process Dynamics

Usage

```
Dynamics(dynamics)
```

Arguments

`dynamics`	`1`, `2`, or `3`. `1` for stable reciprocal regulation, `2` for escalating co-activation, and `3` for adaptive recovery.

See Also

Other Data Generation Functions: `GenData()`

Examples

```
# stable reciprocal regulation,
Dynamics(dynamics = 1)
# escalating co-activation
Dynamics(dynamics = 2)
# adaptive recovery
Dynamics(dynamics = 3)
```

FitDTVAR *Fit the Model using the fitDTVARMxID Package*

Description

The function fits the model using the fitDTVARMxID::fitDTVARMxID package.

Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

Arguments

`data`	R object. Output of the `GenData()` function.
`ncores`	Positive integer. Number of cores to use.
`seed`	Integer. Random seed.

See Also

Other Model Fitting Functions: `FitMLVAR()`, `FitMetaVAR()`, `FitMplus()`, `MplusInput()`

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

`FitMetaVAR` *Multivariate Meta-Analysis using the metaVAR Package*

Description

The function performs multivariate meta-snalysis using the metaVAR::metaVAR package.

Usage

```
FitMetaVAR(fit, ncores = NULL)
```

Arguments

`fit`	R object. Output of the `FitDTVAR()` function.
`ncores`	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMLVAR()`, `FitMplus()`, `MplusInput()`

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
pooled <- FitMetaVAR(
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)
print(pooled)
coef(pooled)
vcov(pooled)

## End(Not run)
```

FitMLVAR *Fit the Model using the mlVAR Package*

Description

The function fits the model using the mlVAR::mlVAR package.

Usage

```
FitMLVAR(data, ncores = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: FitDTVAR(), FitMetaVAR(), FitMplus(), MplusInput()

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMLVAR(data = data)
print(fit)
summary(fit)

## End(Not run)
```

FitMplus *Fit the Model using Mplus*

Description

The function fits the model using Mplus.

Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 60000L,
  fscores = NULL,
  plot = FALSE,
  default_priors = FALSE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL
)
```

Arguments

`data`	R object. Output of the `GenData()` function.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot` = `TRUE`, add `PLOT: TYPE` = `PLOT3;` to `Mplus` input file.
`default_priors`	Logical. If `default_priors` = `TRUE`, use default priors.
`wd`	Character string. Working directory.
`mplus_bin`	Character string. Path to Mplus binary. If `mplus_bin` = `NULL`, the function will try to find the appropriate binary.
`ncores`	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMLVAR()`, `FitMetaVAR()`, `MplusInput()`

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

GenData *Simulate Data*

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid)
```

Arguments

taskid	Positive integer. Task ID.

See Also

Other Data Generation Functions: `Dynamics()`

Examples

```
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)
```

MplusInput *Create Mplus Input File*

Description

The function creates an Mplus input file.

Usage

```
MplusInput(
  dynamics,
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  default_priors,
  ncores = NULL
)
```

Arguments

`dynamics`	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
`fn_data`	Character string. Filename for data file.
`fn_posterior`	Character string. Filename for posterior output.
`fn_factorscores`	Character string. Filename for factor scores output.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
`default_priors`	Logical. If `default_priors = TRUE`, use default priors.
`ncores`	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMLVAR()`, `FitMetaVAR()`, `FitMplus()`

Examples

```
cat(
  MplusInput(
    dynamics = 1,
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 60000L,
    fscores = 1000L,
    plot = TRUE,
    default_priors = FALSE,
    ncores = NULL
  )
)
```

---

params — *Simulation Parameters*

Description

Simulation Parameters

Usage

```
params
```

Format

A dataframe with 25 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occassions.

dynamics Process dynamics. `1` for stable reciprocal regulation, `2` for escalating co-activation, and `3` for adaptive recovery.

Author(s)

Ivan Jacob Agaloos Pesigan

`plot.manmetavar.data` *Plot Method for an Object of Class* `manmetavar.data`

Description

Plot Method for an Object of Class `manmetavar.data`

Plot Method for an Object of Class `manmetavar.mplus`

Usage

```
## S3 method for class 'manmetavar.data'
plot(x, ...)

## S3 method for class 'manmetavar.mplus'
plot(x, what = "posterior", parm = NULL, level = 0.95, burnin = NULL, ...)
```

Arguments

`x`	Object of class `manmetavar.mplus`.
`...`	additional arguments.
`what`	Character string. If `what = "posterior"`, return posterior distribution plots. If `what = "trace"`, return trace plots.
`parm`	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
`level`	the confidence level required.
`burnin`	Integer indicating initial samples to discard. If `burnin = NULL`, do not discard anything.

Author(s)

Ivan Jacob Agaloos Pesigan

print.manmetavar.data *Print Method for an Object of Class* manmetavar.data

Description

Print Method for an Object of Class manmetavar.data

Print Method (FitDTVAR)

Print Method (FitMetaVAR)

Print Method (FitMLVAR)

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)
```

Arguments

`x`	Object of class `manmetavar.mlvar`.
`...`	additional arguments.
`means`	Logical. If `means = TRUE`, return means. Otherwise, the function returns raw estimates.
`alpha`	Numeric vector. Significance level α.
`beta`	Logical. If `beta = TRUE`, include estimates of the `beta` matrix, if available. If `beta = FALSE`, exclude estimates of the `beta` matrix.
`nu`	Logical. If `nu = TRUE`, include estimates of the `nu` vector, if available. If `nu = FALSE`, exclude estimates of the `nu` vector.
`psi`	Logical. If `psi = TRUE`, include estimates of the `psi` matrix, if available. If `psi = FALSE`, exclude estimates of the `psi` matrix.
`theta`	Logical. If `theta = TRUE`, include estimates of the `theta` matrix, if available. If `theta = FALSE`, exclude estimates of the `theta` matrix.
`converged`	Logical. Only include converged cases.
`grad_tol`	Numeric scalar. Tolerance for the maximum absolute gradient if `converged = TRUE`.
`hess_tol`	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if `converged = TRUE`.
`vanishing_theta`	Logical. Test for measurement error variance going to zero if `converged = TRUE`.
`theta_tol`	Numeric. Tolerance for vanishing theta test if `converged` and `theta_tol` are `TRUE`.
`digits`	Integer indicating the number of decimal places to display.
`show`	Tables to show.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim *Simulation Replication*

Description

Simulation Replication

Usage

```
Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  metavar,
  mlvar,
  mplus,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`seed`	Integer. Random seed.
`data`	Logical. Simulate data.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaVAR` package.
`mlvar`	Logical. If `mlvar = TRUE`, fit the model using the `mlVAR` package.
`mplus`	Logical. If `mplus = TRUE`, fit the model using DSEM in `Mplus`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
`default_priors`	Logical. If `default_priors = TRUE`, use default priors.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDTVAR — *Simulation Replication - FitDTVAR*

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR — *Simulation Replication - FitMetaVAR*

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitMLVAR` *Simulation Replication - FitMLVAR*

Description

Simulation Replication - FitMLVAR

Usage

```
SimFitMLVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus *Simulation Replication - FitMplus*

Description

Simulation Replication - FitMplus

Usage

```
SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
`default_priors`	Logical. If `default_priors = TRUE`, use default priors.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFN` *Simulation File Name*

Description

Simulation File Name

Usage

```
SimFN(output_type, output_folder, suffix)
```

Arguments

`output_type`	Character string. Output type. Valid values include `"data"`, `"fit-dt-var-mx"`, `"fit-meta-var-mx"`, and `"fit-ml-var"`.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.

Value

Returns a character string file name with the `output_folder` in the OS-specific format.

`SimGenData` *Simulation Replication - GenData*

Description

Simulation Replication - GenData

Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimProj` *Simulation Project Name*

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

`Sum` *Summary*

Description

Summary

Usage

```
Sum(taskid, reps, output_folder, overwrite, integrity, metavar, mlvar, mplus)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaVAR` package.
`mlvar`	Logical. If `mlvar = TRUE`, fit the model using the `mlVAR` package.
`mplus`	Logical. If `mplus = TRUE`, fit the model using DSEM in `Mplus`.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMetaVARLB` *Summary: Likelihood-Based (FitMetaVAR)*

Description

Summary: Likelihood-Based (FitMetaVAR)

Usage

```
SumFitMetaVARLB(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMetaVARNormal` *Summary: Normal Theory (FitMetaVAR)*

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMLVAR — *Summary (FitMLVAR)*

Description

Summary (FitMLVAR)

Usage

```
SumFitMLVAR(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMplus` *Summary (FitMplus)*

Description

Summary (FitMplus)

Usage

```
SumFitMplus(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`summary.manmetavar.data` *Summary Method for an Object of Class* `manmetavar.data`

Description

Summary Method for an Object of Class `manmetavar.data`

Summary Method (FitDTVAR)

Summary Method (FitMetaVAR)

Summary Method (FitMLVAR)

Summary Method (FitMplus)

Usage

```
## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.mlvar'
summary(
  object,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)
```

Arguments

object	Object of class `manmetavar.mplus`.
...	additional arguments.
means	Logical. If `means = TRUE`, return means. Otherwise, the function returns raw estimates.
alpha	Numeric vector. Significance level α.
beta	Logical. If `beta = TRUE`, include estimates of the `beta` matrix, if available. If `beta = FALSE`, exclude estimates of the `beta` matrix.
nu	Logical. If `nu = TRUE`, include estimates of the `nu` vector, if available. If `nu = FALSE`, exclude estimates of the `nu` vector.

psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
digits	Integer indicating the number of decimal places to display.
show	Tables to show.
median	Logical. If median = TRUE, return median of the posterior. If median = FALSE, return mean of the posterior.
burnin	Integer indicating initial samples to discard. If burnin = NULL, do not discard anything.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Returns a matrix of estimates, standard errors, number of posterior samples, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

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vcov.manmetavar.dtvar *Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)*

Description

Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)

Sampling Covariance Matrix of the Parameter Estimates (FitMetaVAR)

Sampling Covariance Matrix of the Parameter Estimates (FitMplus)

Usage

```
## S3 method for class 'manmetavar.dtvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)

## S3 method for class 'manmetavar.metavar'
vcov(object, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, ...)
```

Arguments

object	Object of class manmetavar.mplus.
alpha	Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector.
beta	Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.
nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.

... additional arguments.

Value

Returns a list of sampling variance-covariance matrices.

Returns the sampling variance-covariance matrix of the estimated parameters.

Returns the sampling variance-covariance matrix of the estimated parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

Index